NAMAL UNIVERSITY MIANWALI

Departmet Of Computer Science

Farmer Customer Agricultural Marketplace

Project Proposal

V 1.0

Complex Computing Problem _____Milestone 1 of Semester Project

Course :	CSC-331- Software Engineering
Submitted to:	Ms.Asyia Batool
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REVISION HISTORY

Date	Version	Description	Approved by
9/10/2026	V 1.0	PROPOSAL SUBMISSION	Pending

REQUIREMENT PROVIDER AGREEMENT

This agreement is made between the Sir Bilal and the Project Team for the development of the proposed software project titled:

Project Tittle: **Farmer-Customer Agricultural Market Place**

Sir Bilal will provide the necessary project information, domain knowledge, user needs, and other relevant details required for understanding the proposed system and will review and validate the information documented by the Project Team. The Project Team will accurately document the agreed project scope and develop the system based on the requirements established and approved during the subsequent requirements engineering process. Any significant changes to the agreed project scope will be discussed with the Requirement Provider and Project Team before implementation. Both parties agree to maintain effective communication and cooperation throughout the project.

Requirement Provider

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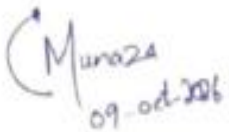


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1. INTRODUCTION:

The Farmer–Customer Agricultural Marketplace is a web-based software system designed to provide a digital platform where farmers can present their agricultural products (crops) directly to potential customers. A farmer will be able to create a farm or crop listing by providing information such as crop name, category, quality, available quantity, expected price, farm location, expected ripening date, ready-to-harvest date, description, and pictures of the farm or crop.

Customers will be able to browse available listings and use filters to find the products they need. The filtering system will organize products into categories such as grains, fruits, and vegetables, with subcategories such as wheat, rice, maize, mango, apple, tomato, potato, and onion. Customers can also view listing details and submit their own price offers to farmers.

The system will allow farmers to view offers from multiple customers, compare the offered prices, and accept an offer. The proposed system belongs to the agricultural marketplace domain and aims to support a more organized interaction between farmers and customers through a single platform.

2. PROBLEM STATEMENT:

Right now, customers who want to buy crops have to travel to different farms or markets. They cannot easily see which crops are available, what the prices are, or what the quality is. This wastes time and travelling cost.

Farmers also have a problem. They do not have an easy way to show their crops to many customers. Most of the time they depend on nearby customers or the local market. So a farmer may not find the best buyer for the crop.

This project will try to solve these problems with a simple online platform, where farmers can post their available crops detail and customers can see and compare them in one place. Customers can search, filter, crop and ask for additional information, and submit price offers. Farmers can view all received offers and select an offer according to their requirements.

3. PROJECT OBJECTIVES:

The objectives of our project are:

1. **User Authentication & Role-Based Access:** To implement a secure authentication mechanism providing role-based access control for Farmers, Customers (Traders), and System Administrators.

2. **Crop Listing Management:** To provide a comprehensive listing module that allows farmers to create, update, and manage crop entries with detailed metadata (crop name, category, subcategory, price, location, harvesting dates, and image attachments).
3. **Taxonomy & Categorization:** To structure crops under a hierarchical taxonomy (Categories and Subcategories) to streamline browsing and navigation.
4. **Automated Inventory Status Tracking:** To automatically update listing statuses (e.g., "Ready for Sale", "Sold") based on offer acceptance and deal completions.
5. **Search & Multi-Criteria Filtering:** To provide a dynamic search engine with multi-criteria filters (category, location, price range, quality, and ready dates) for precise discovery.
6. **Comparative Analysis Tool:** To enable side-by-side comparison of multiple crop listings based on price, quality, quantity, and location metrics.
7. **Offer & Bidding Management:** To facilitate a structured offer management system allowing customers to submit custom price offers and farmers to evaluate, accept, or reject received bids.
8. **In-App Communication Gateway:** To provide an integrated messaging system for direct query resolution and negotiation between farmers and traders regarding listings.
9. **Administrative Governance:** To equip system administrators with moderation tools for user account oversight, content verification, and removal of non-compliant listings.

4. PROJECT SCOPE:

4.1 Farmer Module:

- Farmer registration and login.
- Create a farm/crop listing.
- Enter crop category, crop name, quality, quantity, expected price, location, ripening date, and ready-to-harvest date.
- Upload pictures of the farm or crop and add a detailed description.
- View customer offers for each listing.
- Compare offered prices and identify the highest available offer.
- Accept or reject an offer.

4.2 Customer Module:

- Customer registration and login.
- Browse available farm and crop listings.
- Filter products by category and subcategory, for example Fruits → Mango or Vegetables → Tomato.
- View crop quality, quantity, price, location, pictures, and harvest information.
- Submit a price offer to a farmer.
- Ask for additional information about a listing.

- View the status of submitted offers.

4.3 Administration and System Management

- View, manage, and deactivate registered farmer and customer accounts when required
- Monitor basic system activity and reported information.
- Maintain the system database and application data.

4.4 Project Limitations:

The initial prototype will focus on listing, searching, filtering, customer offers, farmer offer management, and listing updates. Physical delivery, online payment processing, logistics management, crop-quality laboratory verification, and live market-price integration are outside the initial scope unless they are added later as approved requirements.

5. STAKEHOLDER IDENTIFICATION:

The main stakeholders of the proposed system are farmers, customers, and the system administrator. Here customer mean a person(trader) who buy crops from farmers for business or further selling purposes. The project team and requirement provider are also involved in defining, reviewing, and validating the system requirements.

Table 1: Stakeholders and Their Roles in the Proposed System

Stakeholder	Role	Interaction with System
Farmer	Farmer/Seller / Crop Provider	Creates and manages listings, uploads crop details and pictures, views customer offers, and accepts an offer.
Trader	Buyer/Customer	Searches and filters crops, views listings, asks questions, and submits price offers.
Administrator	System Manager	Manages users, categories, listings, and basic system activity.
Project Team	Developers	Analyzes requirements, designs, implements, tests, and maintains the system.
Sir Bilal	Requirement Provider/Academic Adviser	Defines and explains system requirements, provides feedback, and reviews the proposed system to ensure it meets the project requirements.

6. SOFTWARE DEVELOPMENT METHODOLOGY

The project will adopt the Agile Incremental Model (incorporating Scrum practices). While Agile serves as an overarching software development philosophy, the Incremental Model provides the operational strategy by dividing the system into small, manageable feature increments (Farmer Module, Customer Module, Offer Management, etc.).

Each increment will undergo iterative cycles of requirements refinement, interface design, coding, testing, and feedback validation from the Requirement Provider. This ensures early detection of defects, adaptability to evolving requirements, and continuous delivery of functional system components.

Table 6.1: Software Development Plane

Activity	Months	Main Deliverable	Milestone
Requirements Gathering & Analysis	1–2	Software Requirements Specification (SRS)	Requirements approved
System Planning & Architecture	2	System architecture and development plan	Architecture finalized
UI/UX Design	2–3	Wireframes and user interface designs	UI design approved
Database Design	3	ERD and database schema	Database design completed
Frontend Development	3–5	Responsive web pages and user interfaces	Frontend completed
Backend Development & Authentication	4–6	Server-side logic, registration, login, and role-based access	Authentication completed
Farmer Module Development	5–6	Crop listings, farm details, listing management, and image uploads	Farmer module completed
Customer Module Development	6–7	Customer accounts, crop browsing, and listing details	Customer module completed
Search & Filtering Module	7–8	Crop categories, location filters, and search functionality	Search and filtering completed

Activity	Months	Main Deliverable	Milestone
Offer/Bidding Module	8–9	Customer offers, offer management, and farmer offer comparison	Offer workflow completed
Admin Module Development	8–9	User management, listing moderation, and complaint handling	Admin module completed
System Integration	9–10	Integrated frontend, backend, and database	All modules integrated
Testing & Bug Fixing	10–11	Test cases, test reports, security checks, and bug fixes	Major bugs resolved
Deployment & User Acceptance Testing	11	Deployed application and user feedback	Software accepted for release
Final Refinement & Documentation	12	Final working software, user guide, technical documentation, and presentation	Project delivered

7. TOOLS AND TECHNOLOGIES

The following table outlines the technologies selected for the Farmer_Customer Agriculture marketplace web application.

Table 7.1 Proposed Tools and Technologies

#	Tools and Technology	Purpose
1	HTML, CSS	Structure and styling of web pages.
2	JavaScript	Client-side interaction, filtering, forms, and dynamic features.
3	C# / ASP.NET Core	Backend development and application logic (will be used in future).
4	SQL Server SSMS	Storage and management of users, farms, crops, offers, and categories .
5	Figma	UI/UX wireframes and interface prototype design.
6	VS Code	Application development and debugging.
7	GitHub	Version control and team collaboration.

8	Web Browser	Running and testing the web application.
9	Draw.io	For making diagram

8. AI PROMPTS:

[1] ChatGPT (OpenAI), "Our project is a Farmer-Customer Agricultural Marketplace, which will be developed as a web-based application. In this system, farmers can post their crop details, including crop name, category, quantity, quality, price, location, and expected harvesting time. Customers, who are traders and purchase crops for business and resale purposes, can browse available crops and submit price offers to farmers. Farmers can view and compare offers from different customers and select a suitable buyer. Based on these system requirements, explain which technologies and development tools are required to develop this web-based application, including frontend, backend, database, development environment, and other necessary tools," *chatgpt.com*, accessed Sep. 27, 2026.

[2] ChatGPT (OpenAI), "Below are the objectives of our Farmer-Customer Agricultural Marketplace web-based application project. Help me to improve and rewrite these objectives in measurable, clear, and simple terms," *chatgpt.com*, accessed Oct. 5, 2026.

9. Github Repositry Link:

https://github.com/munazabibi130-maker/Farmer-Customer-Agricultural-MarketPlace.git